Ruitang Yang, PhD

Quantitative Scientist | Scientific Computing, Remote Sensing and Probabilistic Modelling

Oslo, Norway | ruitang.yang@geo.uio.no | Portfolio | GitHub | Google Scholar | ORCID

PROFESSIONAL SUMMARY

Quantitative scientist with a background in mathematics, specializing in numerical modelling, data assimilation and satellite remote sensing.

- Led development of a public Python framework for global glacier modelling and applied it regionally in Svalbard to project glacier change through 2100.
- Analyzed three decades of satellite data to quantify regional changes in glacier area, surface elevation, mass and surface velocity in Alaska. Published reusable datasets.

TECHNICAL SKILLS

- **Scientific computing:** Python, Shell, Git, Linux/HPC, MATLAB and C
- **Modelling and inference:** Numerical model integration, joint Bayesian calibration, data assimilation, ensemble simulation and uncertainty quantification
- **Geospatial analysis:** Optical and SAR image processing, DEM generation from stereo imagery, laser altimetry, multi-source data integration, time-series analysis and change detection
- **Geospatial tools**: QGIS, ArcGIS, ENVI and GAMMA

PROFESSIONAL AND RESEARCH EXPERIENCE

- **Researcher |** University of Oslo — Norway | 2022–Present
  - Integrated three numerical models and used joint Bayesian calibration to estimate model parameters from multiple observational datasets.
  - Built reusable Python and shell workflows on Linux/HPC to run ensemble simulations, evaluate model outputs and quantify uncertainty.
  - Co-chair the GlacierMIP4 protocol team, standardizing experiments for international comparisons of global glacier models.
- **Doctoral Researcher |** University of Chinese Academy of Sciences — China | 2016–2022

  Visiting appointments: University of Alaska Fairbanks, US (1.5 yrs); University of Oslo, NO (1 yr)
  - Mapped glacier outlines from optical imagery to measure area change.
  - Generated and differenced elevation models from ASTER stereo imagery to measure surface elevation change.
  - Derived ice velocities using SAR offset tracking and assessed the correlation between seasonal speed variations and modelled runoff.
- **Mathematics Teacher and Class Adviser |** Longxi No.1 Middle School — China | 2011–2016

EDUCATION

- **PhD** in **Physical Geography** | University of Chinese Academy of Sciences — China | 2022
- **MSc** in **Mathematics Education** | Shaanxi Normal University — China | 2015
- **BSc** in **Mathematics and Applied Mathematics** | Shaanxi Normal University — China | 2011

SELECTED SOFTWARE, DATA & PUBLICATIONS

- **Frontal Ablation Modelling Framework** – Led development of a documented Python framework coupling climatic mass balance, ice dynamics, and frontal ablation for observation constrained global glacier simulation and projection. | Software DOI
- **Open Datasets** – Published four reusable datasets comprising three regional datasets derived from remote sensing observations and one model projection dataset, with persistent identifiers and traceable provenance. | Dataset records
- **First-Author Publications** – Authored four peer-reviewed papers covering glacier remote sensing, ice dynamics, ice speed–runoff correlations and coupled numerical modelling with Bayesian calibration. | Google Scholar

PROFESSIONAL ACTIVITIES & RECOGNITION

- **Workshop Organization** – Organizing committee member for CryoDA (Data Assimilation for Cryospheric Applications); the Glacier Modelling workshop at the University of Oslo. 2025
- **Seminar Organization** – Organizing weekly lunch seminars in the Department of Geosciences, University of Oslo, for one semester 2022
- **Research Mentoring** – Co-mentored an MSc student on a thesis in glacier remote sensing 2018
- **Scientific Peer Review** – Reviewer for Journal of Glaciology, Advances in Climate Change Research, and GIScience & Remote Sensing. Evaluator for the AGU Outstanding Student Presentation Award program. 2024-Present
- **Research Recognition** – Article recognized as a Top Cited Article 2022-2023 in Journal of Geophysical Research: Earth Surface 2024
- **Presentation Award** – Best Student Oral Presentation, International Glaciological Society Nordic Branch Symposium 2021

SELECTED FIELD EXPERIENCE

- **Glacier and snow field campaigns in Alaska and China**-ablation measurements, weather station installation, time-lapse monitoring and UAV surveys 2017-2020

LANGUAGES

Chinese (Mandarin) – Native **| English** – Fluent **| Norwegian** – A2 (CEFR)